



Syllabus: Propositional and first order logic.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	0	1	0	1	1	0	1	1	0	0.63	1
2 Marks Count	1	0	0	0	0	0	0	0	0	0.13	1
Total Marks	2	1	0	1	1	0	1	1	0	0.88	2

3.1 First Order Logic (34)

3.1.1 First Order Logic: GATE CSE 1989 | Question: 14a



Symbolize the expression "Every mother loves her children" in predicate logic.

gate1989 descriptive first-order-logic mathematical-logic

Answer key

3.1.2 First Order Logic: GATE CSE 1991 | Question: 15,b



Consider the following first order formula:

$$\left(\begin{array}{c} \forall x \exists y : R(x, y) \\ \wedge \\ \forall x \forall y : (R(x, y) \implies \neg R(y, x)) \\ \wedge \\ \forall x \forall y \forall z : (R(x, y) \wedge R(y, z) \implies R(x, z)) \\ \wedge \\ \forall x : \neg R(x, x) \end{array} \right)$$

Does it have finite models?

Is it satisfiable? If so, give a countable model for it.

gate1991 mathematical-logic first-order-logic descriptive

Answer key

3.1.3 First Order Logic: GATE CSE 1992 | Question: 92,xv



Which of the following predicate calculus statements is/are valid?

- $(\forall(x))P(x) \vee (\forall(x))Q(x) \implies (\forall(x))(P(x) \vee Q(x))$
- $(\exists(x))P(x) \wedge (\exists(x))Q(x) \implies (\exists(x))(P(x) \wedge Q(x))$
- $(\forall(x))(P(x) \vee Q(x)) \implies (\forall(x))P(x) \vee (\forall(x))Q(x)$
- $(\exists(x))(P(x) \vee Q(x)) \implies \sim (\forall(x))P(x) \vee (\exists(x))Q(x)$

gate1992 mathematical-logic normal first-order-logic

Answer key

3.1.4 First Order Logic: GATE CSE 2003 | Question: 32



Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

- $((\forall x)[\alpha] \implies (\forall x)[\beta]) \implies (\forall x)[\alpha \implies \beta]$

- B. $(\forall x)[\alpha] \Rightarrow (\exists x)[\alpha \wedge \beta]$
 C. $((\forall x)[\alpha \vee \beta] \Rightarrow (\exists x)[\alpha]) \Rightarrow (\forall x)[\alpha]$
 D. $(\forall x)[\alpha \Rightarrow \beta] \Rightarrow (((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

gatecse-2003 mathematical-logic first-order-logic normal

Answer key

3.1.5 First Order Logic: GATE CSE 2003 | Question: 33



Consider the following formula and its two interpretations I_1 and I_2 .

$$\alpha: (\forall x) [P_x \Leftrightarrow (\forall y) [Q_{xy} \Leftrightarrow \neg Q_{yy}]] \Rightarrow (\forall x) [\neg P_x]$$

I_1 : Domain: the set of natural numbers

P_x = ' x is a prime number'

Q_{xy} = ' y divides x '

I_2 : same as I_1 except that P_x = ' x is a composite number'.

Which of the following statements is true?

- A. I_1 satisfies α , I_2 does not
 B. I_2 satisfies α , I_1 does not
 C. Neither I_1 nor I_2 satisfies α
 D. Both I_1 and I_2 satisfies α

gatecse-2003 mathematical-logic difficult first-order-logic

Answer key

3.1.6 First Order Logic: GATE CSE 2004 | Question: 23, ISRO2007-32



Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note: taller(x, y) is true if x is taller than y .

- A. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
 B. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
 C. $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
 D. $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

gatecse-2004 mathematical-logic easy isro2007 first-order-logic

Answer key

3.1.7 First Order Logic: GATE CSE 2005 | Question: 41



What is the first order predicate calculus statement equivalent to the following?

"Every teacher is liked by some student"

- A. $\forall(x) [\text{teacher}(x) \rightarrow \exists(y) [\text{student}(y) \rightarrow \text{likes}(y, x)]]$
 B. $\forall(x) [\text{teacher}(x) \rightarrow \exists(y) [\text{student}(y) \wedge \text{likes}(y, x)]]$
 C. $\exists(y) \forall(x) [\text{teacher}(x) \rightarrow [\text{student}(y) \wedge \text{likes}(y, x)]]$
 D. $\forall(x) [\text{teacher}(x) \wedge \exists(y) [\text{student}(y) \rightarrow \text{likes}(y, x)]]$

gatecse-2005 mathematical-logic easy first-order-logic

Answer key

3.1.8 First Order Logic: GATE CSE 2006 | Question: 26



Which one of the first order predicate calculus statements given below correctly expresses the following English statement?

Tigers and lions attack if they are hungry or threatened.

- A. $\forall x[(\text{tiger}(x) \wedge \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$
- B. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \wedge \text{attacks}(x)]$
- C. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow \text{attacks}(x) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x))]$
- D. $\forall x[(\text{tiger}(x) \vee \text{lion}(x)) \rightarrow (\text{hungry}(x) \vee \text{threatened}(x)) \rightarrow \text{attacks}(x)]$

gatecse-2006 mathematical-logic normal first-order-logic

Answer key

3.1.9 First Order Logic: GATE CSE 2007 | Question: 22



Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences **DOES NOT** represent the statement:

“Not every graph is connected”

- A. $\neg \forall x (\text{Graph}(x) \implies \text{Connected}(x))$
- B. $\exists x (\text{Graph}(x) \wedge \neg \text{Connected}(x))$
- C. $\neg \forall x (\neg \text{Graph}(x) \vee \text{Connected}(x))$
- D. $\forall x (\text{Graph}(x) \implies \neg \text{Connected}(x))$

gatecse-2007 mathematical-logic easy first-order-logic

Answer key

3.1.10 First Order Logic: GATE CSE 2008 | Question: 30



Let fsa and pda be two predicates such that $\text{fsa}(x)$ means x is a finite state automaton and $\text{pda}(y)$ means that y is a pushdown automaton. Let equivalent be another predicate such that $\text{equivalent}(a, b)$ means a and b are equivalent. Which of the following first order logic statements represent the following?

Each finite state automaton has an equivalent pushdown automaton

- A. $(\forall x \text{ fsa}(x)) \implies (\exists y \text{ pda}(y) \wedge \text{equivalent}(x, y))$
- B. $\neg \forall y (\exists x \text{ fsa}(x) \implies \text{pda}(y) \wedge \text{equivalent}(x, y))$
- C. $\forall x \exists y (\text{fsa}(x) \wedge \text{pda}(y) \wedge \text{equivalent}(x, y))$
- D. $\forall x \exists y (\text{fsa}(y) \wedge \text{pda}(x) \wedge \text{equivalent}(x, y))$

gatecse-2008 easy mathematical-logic first-order-logic

Answer key

3.1.11 First Order Logic: GATE CSE 2009 | Question: 23



Which one of the following is the most appropriate logical formula to represent the statement?

"Gold and silver ornaments are precious".

The following notations are used:

- $G(x)$: x is a gold ornament
- $S(x)$: x is a silver ornament
- $P(x)$: x is precious

- A. $\forall x (P(x) \implies (G(x) \wedge S(x)))$
- B. $\forall x ((G(x) \wedge S(x)) \implies P(x))$
- C. $\exists x ((G(x) \wedge S(x)) \implies P(x))$
- D. $\forall x ((G(x) \vee S(x)) \implies P(x))$

gatecse-2009 mathematical-logic easy first-order-logic

Answer key

3.1.12 First Order Logic: GATE CSE 2009 | Question: 26



Consider the following well-formed formulae:

- I. $\neg \forall x(P(x))$
- II. $\neg \exists x(P(x))$
- III. $\neg \exists x(\neg P(x))$
- IV. $\exists x(\neg P(x))$

Which of the above are equivalent?

- A. I and III
- B. I and IV
- C. II and III
- D. II and IV

gatecse-2009 mathematical-logic normal first-order-logic

Answer key

3.1.13 First Order Logic: GATE CSE 2010 | Question: 30



Suppose the predicate $F(x, y, t)$ is used to represent the statement that person x can fool person y at time t . Which one of the statements below expresses best the meaning of the formula,

$$\forall x \exists y \exists t (\neg F(x, y, t))$$

- A. Everyone can fool some person at some time
- B. No one can fool everyone all the time
- C. Everyone cannot fool some person all the time
- D. No one can fool some person at some time

gatecse-2010 mathematical-logic easy first-order-logic

Answer key

3.1.14 First Order Logic: GATE CSE 2011 | Question: 30



Which one of the following options is CORRECT given three positive integers x, y and z , and a predicate

$$P(x) = \neg(x = 1) \wedge \forall y (\exists z (x = y * z) \Rightarrow (y = x) \vee (y = 1))$$

- A. $P(x)$ being true means that x is a prime number
- B. $P(x)$ being true means that x is a number other than 1
- C. $P(x)$ is always true irrespective of the value of x
- D. $P(x)$ being true means that x has exactly two factors other than 1 and x

gatecse-2011 mathematical-logic normal first-order-logic

Answer key

3.1.15 First Order Logic: GATE CSE 2012 | Question: 13



What is the correct translation of the following statement into mathematical logic?

"Some real numbers are rational"

- A. $\exists x(\text{real}(x) \vee \text{rational}(x))$
- B. $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$
- C. $\exists x(\text{real}(x) \wedge \text{rational}(x))$
- D. $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

gatecse-2012 mathematical-logic easy first-order-logic

Answer key

3.1.16 First Order Logic: GATE CSE 2013 | Question: 27



What is the logical translation of the following statement?

"None of my friends are perfect."

- A. $\exists x(F(x) \wedge \neg P(x))$
- B. $\exists x(\neg F(x) \wedge P(x))$
- C. $\exists x(\neg F(x) \wedge \neg P(x))$
- D. $\neg \exists x(F(x) \wedge P(x))$

gatecse-2013 mathematical-logic easy first-order-logic

Answer key

3.1.17 First Order Logic: GATE CSE 2013 | Question: 47



Which one of the following is **NOT** logically equivalent to $\neg \exists x (\forall y (\alpha) \wedge \forall z (\beta))$?

- A. $\forall x (\exists z (\neg \beta) \rightarrow \forall y (\alpha))$
- B. $\forall x (\forall z (\beta) \rightarrow \exists y (\neg \alpha))$
- C. $\forall x (\forall y (\alpha) \rightarrow \exists z (\neg \beta))$
- D. $\forall x (\exists y (\neg \alpha) \rightarrow \exists z (\neg \beta))$

mathematical-logic normal marks-to-all gatecse-2013 first-order-logic

Answer key

3.1.18 First Order Logic: GATE CSE 2014 Set 1 | Question: 1



Consider the statement

"Not all that glitters is gold"

Predicate $\text{glitters}(x)$ is true if x glitters and predicate $\text{gold}(x)$ is true if x is gold. Which one of the following logical formulae represents the above statement?

- A. $\forall x : \text{glitters}(x) \Rightarrow \neg \text{gold}(x)$
- B. $\forall x : \text{gold}(x) \Rightarrow \text{glitters}(x)$
- C. $\exists x : \text{gold}(x) \wedge \neg \text{glitters}(x)$
- D. $\exists x : \text{glitters}(x) \wedge \neg \text{gold}(x)$

gatecse-2014-set1 mathematical-logic first-order-logic

Answer key

3.1.19 First Order Logic: GATE CSE 2014 Set 3 | Question: 53



The CORRECT formula for the sentence, "not all Rainy days are Cold" is

- A. $\forall d (\text{Rainy}(d) \wedge \neg \text{Cold}(d))$
- B. $\forall d (\neg \text{Rainy}(d) \rightarrow \text{Cold}(d))$
- C. $\exists d (\neg \text{Rainy}(d) \rightarrow \text{Cold}(d))$
- D. $\exists d (\text{Rainy}(d) \wedge \neg \text{Cold}(d))$

gatecse-2014-set3 mathematical-logic easy first-order-logic

Answer key

3.1.20 First Order Logic: GATE CSE 2015 Set 2 | Question: 55



Which one of the following well-formed formulae is a tautology?

- A. $\forall x \exists y R(x, y) \leftrightarrow \exists y \forall x R(x, y)$
- B. $(\forall x [\exists y R(x, y) \rightarrow S(x, y)]) \rightarrow \forall x \exists y S(x, y)$
- C. $[\forall x \exists y (P(x, y) \rightarrow R(x, y))] \leftrightarrow [\forall x \exists y (\neg P(x, y) \vee R(x, y))]$
- D. $\forall x \forall y P(x, y) \rightarrow \forall x \forall y P(y, x)$

gatecse-2015-set2 mathematical-logic normal first-order-logic

Answer key

3.1.21 First Order Logic: GATE CSE 2016 Set 2 | Question: 27



Which one of the following well-formed formulae in predicate calculus is **NOT** valid ?

- A. $(\forall x p(x) \implies \forall x q(x)) \implies (\exists x \neg p(x) \vee \forall x q(x))$
- B. $(\exists x p(x) \vee \exists x q(x)) \implies \exists x (p(x) \vee q(x))$
- C. $\exists x (p(x) \wedge q(x)) \implies (\exists x p(x) \wedge \exists x q(x))$
- D. $\forall x (p(x) \vee q(x)) \implies (\forall x p(x) \vee \forall x q(x))$

gatecse-2016-set2 mathematical-logic first-order-logic normal

Answer key

3.1.22 First Order Logic: GATE CSE 2017 Set 1 | Question: 02



Consider the first-order logic sentence $F : \forall x (\exists y R(x, y))$. Assuming non-empty logical domains, which of the

sentences below are *implied* by F ?

- I. $\exists y(\exists x R(x, y))$
- II. $\exists y(\forall x R(x, y))$
- III. $\forall y(\exists x R(x, y))$
- IV. $\neg \exists x(\forall y \neg R(x, y))$

- A. IV only B. I and IV only C. II only D. II and III only

gatecse-2017-set1 mathematical-logic first-order-logic

Answer key

3.1.23 First Order Logic: GATE CSE 2018 | Question: 28



Consider the first-order logic sentence

$$\varphi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \psi(s, t, u, v, w, x, y)$$

where $\psi(s, t, u, v, w, x, y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

- A. There exists at least one model of φ with universe of size less than or equal to 3
- B. There exists no model of φ with universe of size less than or equal to 3
- C. There exists no model of φ with universe size of greater than 7
- D. Every model of φ has a universe of size equal to 7

gatecse-2018 mathematical-logic normal first-order-logic two-marks

Answer key

3.1.24 First Order Logic: GATE CSE 2019 | Question: 35



Consider the first order predicate formula φ :

$$\forall x[(\forall z z|x \Rightarrow ((z = x) \vee (z = 1))) \rightarrow \exists w(w > x) \wedge (\forall z z|w \Rightarrow ((w = z) \vee (z = 1)))]$$

Here $a | b$ denotes that ' a divides b ', where a and b are integers. Consider the following sets:

- $S_1 : \{1, 2, 3, \dots, 100\}$
- $S_2 : \text{Set of all positive integers}$
- $S_3 : \text{Set of all integers}$

Which of the above sets satisfy φ ?

- A. S_1 and S_2 B. S_1 and S_3 C. S_2 and S_3 D. S_1, S_2 and S_3

gatecse-2019 engineering-mathematics discrete-mathematics mathematical-logic first-order-logic two-marks

Answer key

3.1.25 First Order Logic: GATE CSE 2020 | Question: 39



Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x .

- A. $\forall x(p(x) \vee W) \equiv \forall x(p(x)) \vee W$
- B. $\exists x(p(x) \wedge W) \equiv \exists x(p(x)) \wedge W$
- C. $\forall x(p(x) \rightarrow W) \equiv \forall x(p(x)) \rightarrow W$
- D. $\exists x(p(x) \rightarrow W) \equiv \forall x(p(x)) \rightarrow W$

gatecse-2020 first-order-logic mathematical-logic two-marks

Answer key

3.1.26 First Order Logic: GATE CSE 2023 | Question: 16



Geetha has a conjecture about integers, which is of the form

$$\forall x(P(x) \implies \exists yQ(x,y)),$$

where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would *imply* Geetha's conjecture?

- A. $\exists x(P(x) \wedge \forall yQ(x,y))$
- B. $\forall x\forall yQ(x,y)$
- C. $\exists y\forall x(P(x) \implies Q(x,y))$
- D. $\exists x(P(x) \wedge \exists yQ(x,y))$

gatecse-2023 mathematical-logic first-order-logic multiple-selects one-mark

Answer key

3.1.27 First Order Logic: GATE CSE 2025 | Set 1 | Question: 38



Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "Everyone has exactly one mother"?

The meanings of the predicates used are:

- mother (y, x) : y is the mother of x
- noteq (x, y) : x and y are not equal

- A. $\forall x\exists y\exists z(\text{mother}(y, x) \wedge \neg \text{mother}(z, x))$
- B. $\forall x\exists y[\text{mother}(y, x) \wedge \forall z(\text{noteq}(z, y) \rightarrow \neg \text{mother}(z, x))]$
- C. $\forall x\forall y[\text{mother}(y, x) \rightarrow \exists z(\text{mother}(z, x) \wedge \neg \text{noteq}(z, y))]$
- D. $\forall x\exists y[\text{mother}(y, x) \wedge \neg \exists z(\text{noteq}(z, y) \wedge \text{mother}(z, x))]$

gatecse2025-set1 mathematical-logic first-order-logic multiple-selects two-marks

Answer key

3.1.28 First Order Logic: GATE CSE 2025 | Set 2 | Question: 5



Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?

- A. $(P(0) \wedge (\forall x[P(x) \implies P(x+1)])) \implies \forall xP(x)$
- B. $(P(0) \wedge (\forall x[P(x) \implies P(x-1)])) \implies \forall xP(x)$
- C. $(P(1000) \wedge (\forall x[P(x) \implies P(x-1)])) \implies \forall xP(x)$
- D. $(P(1000) \wedge (\forall x[P(x) \implies P(x+1)])) \implies \forall xP(x)$

gatecse2025-set2 mathematical-logic first-order-logic one-mark

Answer key

3.1.29 First Order Logic: GATE IT 2004 | Question: 3



Let $a(x, y)$, $b(x, y)$ and $c(x, y)$ be three statements with variables x and y chosen from some universe. Consider the following statement:

$$(\exists x)(\forall y)[(a(x, y) \wedge b(x, y)) \wedge \neg c(x, y)]$$

Which one of the following is its equivalent?

- A. $(\forall x)(\exists y)[(a(x, y) \vee b(x, y)) \rightarrow c(x, y)]$
- B. $(\exists x)(\forall y)[(a(x, y) \vee b(x, y)) \wedge \neg c(x, y)]$
- C. $\neg(\forall x)(\exists y)[(a(x, y) \wedge b(x, y)) \rightarrow c(x, y)]$
- D. $\neg(\forall x)(\exists y)[(a(x, y) \vee b(x, y)) \rightarrow c(x, y)]$

gateit-2004 mathematical-logic normal discrete-mathematics first-order-logic

Answer key

3.1.30 First Order Logic: GATE IT 2005 | Question: 36



Let $P(x)$ and $Q(x)$ be arbitrary predicates. Which of the following statements is always **TRUE**?

- A. $((\forall x (P(x) \vee Q(x)))) \implies ((\forall x P(x)) \vee (\forall x Q(x)))$
- B. $(\forall x (P(x) \implies Q(x))) \implies ((\forall x P(x)) \implies (\forall x Q(x)))$
- C. $(\forall x (P(x) \implies \forall x (Q(x)))) \implies (\forall x (P(x) \implies Q(x)))$
- D. $(\forall x (P(x) \Leftrightarrow (\forall x (Q(x)))) \implies (\forall x (P(x) \Leftrightarrow Q(x)))$

gateit-2005 mathematical-logic first-order-logic normal

Answer key

3.1.31 First Order Logic: GATE IT 2006 | Question: 21



Consider the following first order logic formula in which R is a binary relation symbol.

$$\forall x \forall y (R(x, y) \implies R(y, x))$$

The formula is

- A. satisfiable and valid
- B. satisfiable and so is its negation
- C. unsatisfiable but its negation is valid
- D. satisfiable but its negation is unsatisfiable

gateit-2006 mathematical-logic normal first-order-logic

Answer key

3.1.32 First Order Logic: GATE IT 2007 | Question: 21



Which one of these first-order logic formulae is valid?

- A. $\forall x (P(x) \implies Q(x)) \implies (\forall x P(x) \implies \forall x Q(x))$
- B. $\exists x (P(x) \vee Q(x)) \implies (\exists x P(x) \implies \exists x Q(x))$
- C. $\exists x (P(x) \wedge Q(x)) \iff (\exists x P(x) \wedge \exists x Q(x))$
- D. $\forall x \exists y P(x, y) \implies \exists y \forall x P(x, y)$

gateit-2007 mathematical-logic normal first-order-logic

Answer key

3.1.33 First Order Logic: GATE IT 2008 | Question: 21



Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first order formula with x as a free variable, and β is a first order formula with no free variable.

- A. $[\beta \rightarrow (\exists x, \alpha(x))] \rightarrow [\forall x, \beta \rightarrow \alpha(x)]$
- B. $[\exists x, \beta \rightarrow \alpha(x)] \rightarrow [\beta \rightarrow (\forall x, \alpha(x))]$
- C. $[(\exists x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$
- D. $[(\forall x, \alpha(x)) \rightarrow \beta] \rightarrow [\forall x, \alpha(x) \rightarrow \beta]$

gateit-2008 first-order-logic normal

Answer key

3.1.34 First Order Logic: GATE IT 2008 | Question: 22



Which of the following is the negation of $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, y))]$

- A. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, y))]$
- B. $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg y))]$
- C. $[\forall x, \neg \alpha \rightarrow (\exists y, \neg \beta \rightarrow (\forall u, \exists v, \neg y))]$
- D. $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg y))]$

gateit-2008 mathematical-logic normal first-order-logic

Answer key

3.2.1 Logical Reasoning: GATE CSE 2012 | Question: 1



Consider the following logical inferences.

I_1 : If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.

I_2 : If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.
Which of the following is **TRUE**?

- A. Both I_1 and I_2 are correct inferences
- B. I_1 is correct but I_2 is not a correct inference
- C. I_1 is not correct but I_2 is a correct inference
- D. Both I_1 and I_2 are not correct inferences

gatecse-2012 mathematical-logic easy logical-reasoning

Answer key

3.2.2 Logical Reasoning: GATE CSE 2015 Set 2 | Question: 3



Consider the following two statements.

- S_1 : If a candidate is known to be corrupt, then he will not be elected
- S_2 : If a candidate is kind, he will be elected

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

- A. If a person is known to be corrupt, he is kind
- B. If a person is not known to be corrupt, he is not kind
- C. If a person is kind, he is not known to be corrupt
- D. If a person is not kind, he is not known to be corrupt

gatecse-2015-set2 mathematical-logic normal logical-reasoning

Answer key

3.2.3 Logical Reasoning: GATE CSE 2015 Set 3 | Question: 24



In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking the person replies the following

"The result of the toss is head if and only if I am telling the truth"

Which of the following options is correct?

- A. The result is head
- B. The result is tail
- C. If the person is of Type 2, then the result is tail
- D. If the person is of Type 1, then the result is tail

gatecse-2015-set3 mathematical-logic difficult logical-reasoning

Answer key

3.3.1 Propositional Logic: GATE CSE 1987 | Question: 10e



Show that the conclusion $(r \rightarrow q)$ follows from the premises: $p, (p \rightarrow q) \vee (p \wedge (r \rightarrow q))$

gate1987 mathematical-logic propositional-logic proof descriptive

Answer key

3.3.2 Propositional Logic: GATE CSE 1988 | Question: 2vii



Define the validity of a well-formed formula(wff)?

gate1988 descriptive mathematical-logic propositional-logic

Answer key

3.3.3 Propositional Logic: GATE CSE 1989 | Question: 3-v



Which of the following well-formed formulas are equivalent?

- A. $P \rightarrow Q$ B. $\neg Q \rightarrow \neg P$
C. $\neg P \vee Q$ D. $\neg Q \rightarrow P$

gate1989 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.4 Propositional Logic: GATE CSE 1990 | Question: 3-x



Indicate which of the following well-formed formulae are valid:

- A. $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \Rightarrow (P \Rightarrow R)$
B. $(P \Rightarrow Q) \Rightarrow (\neg P \Rightarrow \neg Q)$
C. $(P \wedge (\neg P \vee \neg Q)) \Rightarrow Q$
D. $(P \Rightarrow R) \vee (Q \Rightarrow R) \Rightarrow ((P \vee Q) \Rightarrow R)$

gate1990 normal mathematical-logic propositional-logic multiple-selects

Answer key

3.3.5 Propositional Logic: GATE CSE 1991 | Question: 03,xii



If F_1 , F_2 and F_3 are propositional formulae such that $F_1 \wedge F_2 \rightarrow F_3$ and $F_1 \wedge F_2 \rightarrow \sim F_3$ are both tautologies, then which of the following is true:

- A. Both F_1 and F_2 are tautologies B. The conjunction $F_1 \wedge F_2$ is not satisfiable
C. Neither is tautologous D. Neither is satisfiable
E. None of the above

gate1991 mathematical-logic normal propositional-logic multiple-selects

Answer key

3.3.6 Propositional Logic: GATE CSE 1992 | Question: 02,xvi



Which of the following is/are a tautology?

- A. $a \vee b \rightarrow b \wedge c$ B. $a \wedge b \rightarrow b \vee c$
C. $a \vee b \rightarrow (b \rightarrow c)$ D. $a \rightarrow b \rightarrow (b \rightarrow c)$

gate1992 mathematical-logic easy propositional-logic multiple-selects

Answer key

3.3.7 Propositional Logic: GATE CSE 1992 | Question: 15.a



Use Modus ponens ($A, A \rightarrow B \vdash B$) or resolution to show that the following set is inconsistent:

1. $Q(x) \rightarrow P(x) \vee \sim R(a)$
2. $R(a) \vee \sim Q(a)$
3. $Q(a)$
4. $\sim P(y)$

where x and y are universally quantified variables, a is a constant and P, Q, R are monadic predicates.

gate1992 normal mathematical-logic propositional-logic descriptive

Answer key

3.3.8 Propositional Logic: GATE CSE 1993 | Question: 18



Show that proposition C is a logical consequence of the formula

$$A \wedge (A \rightarrow (B \vee C)) \wedge (B \rightarrow \neg A)$$

using truth tables.

gate1993 mathematical-logic normal propositional-logic proof descriptive

Answer key

3.3.9 Propositional Logic: GATE CSE 1993 | Question: 8.2



The proposition $p \wedge (\sim p \vee q)$ is:

- A. a tautology
- B. logically equivalent to $p \wedge q$
- C. logically equivalent to $p \vee q$
- D. a contradiction
- E. none of the above

gate1993 mathematical-logic easy propositional-logic

Answer key

3.3.10 Propositional Logic: GATE CSE 1994 | Question: 3.13



Let p and q be propositions. Using only the Truth Table, decide whether

- $p \iff q$ does not imply $p \rightarrow \neg q$

is **True** or **False**.

gate1994 mathematical-logic normal propositional-logic true-false

Answer key

3.3.11 Propositional Logic: GATE CSE 1995 | Question: 13



Obtain the principal (canonical) conjunctive normal form of the propositional formula

$$(p \wedge q) \vee (\neg q \wedge r)$$

where $\wedge$ is logical and, $\vee$ is inclusive or and $\neg$ is negation.

gate1995 mathematical-logic propositional-logic normal descriptive

Answer key

3.3.12 Propositional Logic: GATE CSE 1995 | Question: 2.19



If the proposition $\neg p \rightarrow q$ is true, then the truth value of the proposition $\neg p \vee (p \rightarrow q)$, where $\neg$ is negation, $\vee$ is inclusive OR and $\rightarrow$ is implication, is

- A. True
- B. Multiple Values
- C. False
- D. Cannot be determined

gate1995 mathematical-logic normal propositional-logic

Answer key

3.3.13 Propositional Logic: GATE CSE 1996 | Question: 2.3



Which of the following is NOT True?

(Read $\wedge$ as AND, $\vee$ as OR, $\neg$ as NOT, $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication)

- A. $((x \rightarrow y) \wedge x) \rightarrow y$
- B. $((\neg x \rightarrow y) \wedge (\neg x \rightarrow \neg y)) \rightarrow x$
- C. $(x \rightarrow (x \vee y))$
- D. $((x \vee y) \leftrightarrow (\neg x \rightarrow \neg y))$

gate1996 mathematical-logic normal propositional-logic

Answer key

3.3.14 Propositional Logic: GATE CSE 1997 | Question: 3.2



Which of the following propositions is a tautology?

- A. $(p \vee q) \rightarrow p$
C. $p \vee (p \rightarrow q)$
- B. $p \vee (q \rightarrow p)$
D. $p \rightarrow (p \rightarrow q)$

gate1997 mathematical-logic easy propositional-logic

Answer key

3.3.15 Propositional Logic: GATE CSE 1998 | Question: 1.5



What is the converse of the following assertion?

- I stay only if you go

- A. I stay if you go
C. If you do not go then I do not stay
- B. If I stay then you go
D. If I do not stay then you go

gate1998 mathematical-logic easy propositional-logic

Answer key

3.3.16 Propositional Logic: GATE CSE 1999 | Question: 14



Show that the formula $[(\sim p \vee q) \Rightarrow (q \Rightarrow p)]$ is not a tautology.

Let A be a tautology and B any other formula. Prove that $(A \vee B)$ is a tautology.

gate1999 mathematical-logic normal propositional-logic proof descriptive

Answer key

3.3.17 Propositional Logic: GATE CSE 2000 | Question: 2.7



Let a, b, c, d be propositions. Assume that the equivalence $a \Leftrightarrow (b \vee \neg b)$ and $b \Leftrightarrow c$ hold. Then the truth-value of the formula $(a \wedge b) \rightarrow (a \wedge c) \vee d$ is always

- A. True
C. Same as the truth-value of b
- B. False
D. Same as the truth-value of d

gatecse-2000 mathematical-logic normal propositional-logic

Answer key

3.3.18 Propositional Logic: GATE CSE 2001 | Question: 1.3



Consider two well-formed formulas in propositional logic

$$F_1 : P \Rightarrow \neg P \quad F_2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$$

Which one of the following statements is correct?

- A. F_1 is satisfiable, F_2 is valid
C. F_1 is unsatisfiable, F_2 is valid
- B. F_1 unsatisfiable, F_2 is satisfiable
D. F_1 and F_2 are both satisfiable

gatecse-2001 mathematical-logic easy propositional-logic

Answer key

3.3.19 Propositional Logic: GATE CSE 2002 | Question: 1.8



"If X then Y unless Z " is represented by which of the following formulas in propositional logic? (" $\neg$ " is negation, " $\wedge$ " is conjunction, and " $\rightarrow$ " is implication)

- A. $(X \wedge \neg Z) \rightarrow Y$
C. $X \rightarrow (Y \wedge \neg Z)$
- B. $(X \wedge Y) \rightarrow \neg Z$
D. $(X \rightarrow Y) \wedge \neg Z$

gatecse-2002 mathematical-logic normal propositional-logic

Answer key

3.3.20 Propositional Logic: GATE CSE 2002 | Question: 5b



Determine whether each of the following is a tautology, a contradiction, or neither (" $\vee$ " is disjunction, " $\wedge$ " is conjunction, " $\rightarrow$ " is implication, " $\neg$ " is negation, and " $\leftrightarrow$ " is biconditional (if and only if)).

1. $A \leftrightarrow (A \vee A)$
2. $(A \vee B) \rightarrow B$
3. $A \wedge (\neg(A \vee B))$

gatecse-2002 mathematical-logic easy descriptive propositional-logic

Answer key 

3.3.21 Propositional Logic: GATE CSE 2003 | Question: 72



The following resolution rule is used in logic programming.
Derive clause $(P \vee Q)$ from clauses $(P \vee R), (Q \vee \neg R)$
Which of the following statements related to this rule is FALSE?

- A. $((P \vee R) \wedge (Q \vee \neg R)) \Rightarrow (P \vee Q)$ is logically valid
- B. $(P \vee Q) \Rightarrow ((P \vee R) \wedge (Q \vee \neg R))$ is logically valid
- C. $(P \vee Q)$ is satisfiable if and only if $(P \vee R) \wedge (Q \vee \neg R)$ is satisfiable
- D. $(P \vee Q) \Rightarrow \text{FALSE}$ if and only if both P and Q are unsatisfiable

gatecse-2003 mathematical-logic normal propositional-logic

Answer key 

3.3.22 Propositional Logic: GATE CSE 2004 | Question: 70



The following propositional statement is $(P \Rightarrow (Q \vee R)) \Rightarrow ((P \wedge Q) \Rightarrow R)$

- A. satisfiable but not valid
- B. valid
- C. a contradiction
- D. None of the above

gatecse-2004 mathematical-logic normal propositional-logic

Answer key 

3.3.23 Propositional Logic: GATE CSE 2005 | Question: 40



Let P, Q , and R be three atomic propositional assertions. Let X denote $(P \vee Q) \rightarrow R$ and Y denote $(P \rightarrow R) \vee (Q \rightarrow R)$. Which one of the following is a tautology?

- A. $X \equiv Y$
- B. $X \rightarrow Y$
- C. $Y \rightarrow X$
- D. $\neg Y \rightarrow X$

gatecse-2005 mathematical-logic propositional-logic normal

Answer key 

3.3.24 Propositional Logic: GATE CSE 2006 | Question: 27



Consider the following propositional statements:

- $P_1 : ((A \wedge B) \rightarrow C) \equiv ((A \rightarrow C) \wedge (B \rightarrow C))$
- $P_2 : ((A \vee B) \rightarrow C) \equiv ((A \rightarrow C) \vee (B \rightarrow C))$

Which one of the following is true?

- A. P_1 is a tautology, but not P_2
- B. P_2 is a tautology, but not P_1
- C. P_1 and P_2 are both tautologies
- D. Both P_1 and P_2 are not tautologies

gatecse-2006 mathematical-logic normal propositional-logic

Answer key 

3.3.25 Propositional Logic: GATE CSE 2008 | Question: 31



P and Q are two propositions. Which of the following logical expressions are equivalent?

- I. $P \vee \neg Q$
- II. $\neg(\neg P \wedge Q)$
- III. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
- IV. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

- A. Only I and II
- B. Only I, II and III

C. Only I, II and IV

D. All of I, II, III and IV

gatecse-2008 normal mathematical-logic propositional-logic

Answer key

3.3.26 Propositional Logic: GATE CSE 2009 | Question: 24



The binary operation $\square$ is defined as follows

P	Q	$P \square Q$
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

A. $\neg Q \square \neg P$

B. $P \square \neg Q$

C. $\neg P \square Q$

D. $\neg P \square \neg Q$

gatecse-2009 mathematical-logic easy propositional-logic

Answer key

3.3.27 Propositional Logic: GATE CSE 2014 Set 1 | Question: 53



Which one of the following propositional logic formulas is TRUE when exactly two of p, q and r are TRUE?

- A. $((p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
B. $(\sim(p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
C. $((p \rightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$
D. $(\sim(p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

gatecse-2014-set1 mathematical-logic normal propositional-logic

Answer key

3.3.28 Propositional Logic: GATE CSE 2014 Set 2 | Question: 53



Which one of the following Boolean expressions is NOT a tautology?

- A. $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
B. $(a \rightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
C. $(a \wedge b \wedge c) \rightarrow (c \vee a)$
D. $a \rightarrow (b \rightarrow a)$

gatecse-2014-set2 mathematical-logic propositional-logic normal

Answer key

3.3.29 Propositional Logic: GATE CSE 2014 Set 3 | Question: 1



Consider the following statements:

- P: Good mobile phones are not cheap
- Q: Cheap mobile phones are not good

L: P implies Q

M: Q implies P

N: P is equivalent to Q

Which one of the following about L, M, and N is CORRECT?

A. Only L is TRUE.

B. Only M is TRUE.

C. Only N is TRUE.

D. L, M and N are TRUE.

gatecse-2014-set3 mathematical-logic easy propositional-logic

Answer key

3.3.30 Propositional Logic: GATE CSE 2015 Set 1 | Question: 14



Which one of the following is NOT equivalent to $p \leftrightarrow q$?

- A. $(\neg p \vee q) \wedge (p \vee \neg q)$
- B. $(\neg p \vee q) \wedge (q \rightarrow p)$
- C. $(\neg p \wedge q) \vee (p \wedge \neg q)$
- D. $(\neg p \wedge \neg q) \vee (p \wedge q)$

gatecse-2015-set1 mathematical-logic easy propositional-logic

Answer key

3.3.31 Propositional Logic: GATE CSE 2016 Set 1 | Question: 1



Let p, q, r, s represents the following propositions.

- $p : x \in \{8, 9, 10, 11, 12\}$
- $q : x$ is a composite number.
- $r : x$ is a perfect square.
- $s : x$ is a prime number.

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____.

gatecse-2016-set1 mathematical-logic normal numerical-answers propositional-logic

Answer key

3.3.32 Propositional Logic: GATE CSE 2016 Set 2 | Question: 01



Consider the following expressions:

- i. *false*
- ii. Q
- iii. *true*
- iv. $P \vee Q$
- v. $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____.

gatecse-2016-set2 mathematical-logic normal numerical-answers propositional-logic

Answer key

3.3.33 Propositional Logic: GATE CSE 2017 Set 1 | Question: 01



The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

- I. $p \Rightarrow q$
- II. $q \Rightarrow p$
- III. $(\neg q) \vee p$
- IV. $(\neg p) \vee q$

- A. I only
- C. II only

- B. I and IV only
- D. II and III only

gatecse-2017-set1 mathematical-logic propositional-logic easy

Answer key

3.3.34 Propositional Logic: GATE CSE 2017 Set 1 | Question: 29



Let p, q and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

- A. a tautology
- C. always TRUE when p is FALSE

- B. a contradiction
- D. always TRUE when q is TRUE

Answer key

3.3.35 Propositional Logic: GATE CSE 2017 Set 2 | Question: 11



Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by

- A. $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$ B. $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
 C. $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$ D. $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

gatecse-2017-set2 mathematical-logic propositional-logic

Answer key

3.3.36 Propositional Logic: GATE CSE 2021 Set 1 | Question: 7



Let p and q be two propositions. Consider the following two formulae in propositional logic.

- $S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$
- $S_2 : q \rightarrow (\neg p \wedge (p \vee q))$

Which one of the following choices is correct?

- A. Both S_1 and S_2 are tautologies.
 B. S_1 is a tautology but S_2 is not a tautology
 C. S_1 is not a tautology but S_2 is a tautology
 D. Neither S_1 nor S_2 is a tautology

gatecse-2021-set1 mathematical-logic propositional-logic one-mark

Answer key

3.3.37 Propositional Logic: GATE CSE 2021 Set 2 | Question: 15



Choose the correct choice(s) regarding the following propositional logic assertion S :

$$S : ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$$

- A. S is neither a tautology nor a contradiction
 B. S is a tautology
 C. S is a contradiction
 D. The antecedent of S is logically equivalent to the consequent of S

gatecse-2021-set2 multiple-selects mathematical-logic propositional-logic one-mark

Answer key

3.3.38 Propositional Logic: GATE CSE 2024 | Set 2 | Question: 2



Let p and q be the following propositions:

p : Fail grade can be given.

q : Student scores more than 50% marks.

Consider the statement: "Fail grade cannot be given when student scores more than 50% marks."

Which one of the following is the CORRECT representation of the above statement in propositional logic?

- A. $q \rightarrow \neg p$ B. $q \rightarrow p$
 C. $p \rightarrow q$ D. $\neg p \rightarrow q$

gatecse2024-set2 mathematical-logic propositional-logic one-mark

Answer key

3.3.39 Propositional Logic: GATE DS&AI 2024 | Question: 19



Let x and y be two propositions. Which of the following statements is a tautology /are tautologies?

- A. $(\neg x \wedge y) \Rightarrow (y \Rightarrow x)$

- B. $(x \wedge \neg y) \Rightarrow (\neg x \Rightarrow y)$
 C. $(\neg x \wedge y) \Rightarrow (\neg x \Rightarrow y)$
 D. $(x \wedge \neg y) \Rightarrow (y \Rightarrow x)$

gate-ds-ai-2024 mathematical-logic propositional-logic multiple-selects one-mark

Answer key

3.3.40 Propositional Logic: GATE IT 2004 | Question: 31



Let p, q, r and s be four primitive statements. Consider the following arguments:

- $P : [(\neg p \vee q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (\neg s \rightarrow q)$
- $Q : [(\neg p \wedge q) \wedge [q \rightarrow (p \rightarrow r)]] \rightarrow \neg r$
- $R : [(q \wedge r) \rightarrow p] \wedge (\neg q \vee p) \rightarrow r$
- $S : [p \wedge (p \rightarrow r) \wedge (q \vee \neg r)] \rightarrow q$

Which of the above arguments are valid?

- A. P and Q only B. P and R only C. P and S only D. P, Q, R and S

gateit-2004 mathematical-logic normal propositional-logic

Answer key

Answer Keys

3.1.1	N/A	3.1.2	N/A	3.1.3	A	3.1.4	D	3.1.5	D
3.1.6	D	3.1.7	B	3.1.8	D	3.1.9	D	3.1.10	X
3.1.11	D	3.1.12	B	3.1.13	B	3.1.14	A	3.1.15	C
3.1.16	D	3.1.17	X	3.1.18	D	3.1.19	D	3.1.20	C
3.1.21	D	3.1.22	B	3.1.23	A	3.1.24	C	3.1.25	C
3.1.26	B;C	3.1.27	B;D	3.1.28	A	3.1.29	C	3.1.30	B
3.1.31	B	3.1.32	A	3.1.33	C	3.1.34	D	3.2.1	B
3.2.2	C	3.2.3	A	3.3.1	N/A	3.3.2	N/A	3.3.3	A;B;C
3.3.4	A	3.3.5	B	3.3.6	B	3.3.7	N/A	3.3.8	N/A
3.3.9	B	3.3.10	True	3.3.11	N/A	3.3.12	D	3.3.13	D
3.3.14	C	3.3.15	A	3.3.16	N/A	3.3.17	A	3.3.18	A
3.3.19	A	3.3.20	N/A	3.3.21	B	3.3.22	A	3.3.23	B
3.3.24	D	3.3.25	B	3.3.26	B	3.3.27	B	3.3.28	B
3.3.29	D	3.3.30	C	3.3.31	11	3.3.32	4	3.3.33	D
3.3.34	D	3.3.35	A	3.3.36	B	3.3.37	B;D	3.3.38	A
3.3.39	B;C;D	3.3.40	C						